

Octavio Villalobos

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EDUCATION

University of California, Santa Cruz

Expected June 2026

Master of Science in Computer Science — GPA: 3.85/4.00

Santa Cruz, CA

- Relevant Coursework: Artificial Intelligence, Machine Learning, Computer Architecture, Computer Networks, Computer Security, Analysis of Algorithms, GameAI, Foundations of Deep Learning, NLP, Data Science

University of California, Santa Cruz

June 2024

Bachelor of Science in Computer Science, With Honors — GPA: 3.60/4.00

Santa Cruz, CA

SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, TypeScript, C#, SQL, HTML/CSS, Angular

Front-end Development: React.js, React Native (Expo), Tailwind CSS, HTML5

Backend & Databases: FastAPI, Express.js, Node.js, RESTful APIs, PostgreSQL, Supabase

AI/ML & Generative AI: PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, Hugging Face, OpenAI API, Claude API, LoRA, QLoRA, Ollama, ChromaDB, MLflow, Weights & Biases (W&B), Pandas, Matplotlib, Seaborn

Cloud & Infrastructure: Google Cloud Platform (GCP), AWS, Docker, Kubernetes, Linux

DevOps & Testing: Git, GitHub Actions, CI/CD Pipelines, Jenkins, Pytest, Jest

Concepts: Machine Learning, Deep Learning, NLP, Reinforcement Learning, Multi-Agent Systems, RAG, LLM Fine-tuning, Prompt Engineering, Data Pipelines, HDBSCAN, BM25, Dynamic Query Generation, Neural Networks, Large Language Models, Agentic Workflows

EXPERIENCE

Research Project Lead @ Sloth Lab – UCSC

Oct 2025 – Present

Santa Cruz, CA

- Led a 3-person team extending the Lyra opinion dynamics framework into a scalable multi-agent simulation system for personality-driven NPC interactions and emergent social behavior
- Designed agent architectures with 9-dimensional state tracking, integrating **prospect-theory-based** persuasion modeling and 7 adaptive interaction behaviors to simulate realistic non-optimal social dynamics
- Implemented per-agent **Actor-Critic (A2C)** reinforcement-learning policies with modular reward shaping and archetype-specific behavioral constraints, preserving divergent interaction patterns across 5 personality archetypes
- Developed clustering and behavioral-analysis pipelines using **HDBSCAN**, **Jenks Natural Breaks**, and **MLflow** experiment tracking, validating emergent interaction patterns across 1,000+ large-scale simulation runs
- Co-authored a research paper submitted to **AIIDE 2026** (under review) exploring emergent social behavior, conversational dynamics, reinforcement learning, and multi-agent interaction systems

AI Engineer Intern @ Silicon Valley Certification Hub

June 2025 – Present

Palo Alto, CA

- Developed and deployed an **LLM-powered** coaching assistant platform using **Flask** and **AWS**, integrating production web services and model inference pipelines into an existing certification training ecosystem
- Built a RAG-based retrieval pipeline leveraging certification content to generate executive-style scenario assessments and context-aware coaching feedback
- Engineered rubric-based scoring and feedback pipelines using structured **prompt engineering**, configurable grading curves, and multi-criteria evaluation logic to assess answer relevance, ethical risk coverage, mitigation strategies, and actionability
- Fine-tuned **Gemma** and **LLaMA** models using **LoRA**-based parameter-efficient techniques, improving factual alignment by 42.3%, response consistency by 25.1%, and contextual relevance 33.1% for certification coaching
- Implemented role-conditioned scenario generation and optional, optional **ElevenLabs** text-to-speech integration, supporting realistic AI interview simulations across 12 industries

Teaching Assistant @ University of California, Santa Cruz

Sep 2024 – Present

Santa Cruz, CA

- Delivered lectures on databases, AI, LLMs, and technical writing to 150+ students across 4 UCSC courses
- Led weekly discussion sections for 30 students, developing instructional materials and hands-on lesson plans
- Designed coursework, assignments, and exams across 6 courses to improve alignment between lectures and assessments

PROJECTS

FormulaForecast | *Python, Pandas, NumPy, PyTorch, scikit-learn*

Nov 2025 – Jan 2026

- Developed a **deep learning** framework in **PyTorch** for predicting Formula 1 top-10 race finishing positions, leveraging historical race telemetry and driver-performance data from the **FastF1 API**
- Engineered 9 domain-specific implementing **StandardScaler** normalization, **3-sigma outlier clipping**, and filtering pipelines to remove DNFs and high-variance race incidents
- Designed and trained a 3-layer **MLP neural network (128→64→32)** using **ReLU** activations, **Dropout**, **Batch Normalization**, and **He/Kaiming initialization**
- Achieved an MAE of 1.585 and 88.7% within-3 prediction accuracy on unseen 2025 race data while reducing prediction error by 41% compared to baseline full-grid neural network models

Insight Deck | *Python, React, Flask, LangChain, Vector DB, ChromaDB, RAG*

Apr 2025 – Dec 2025

- Co-developed a time-aware hybrid retrieval pipeline combining **BM25**, **ChromaDB** vector search, and metadata-aware reranking, improving form-type retrieval accuracy across multi-year financial corpora
- Orchestrated a multi-agent **RAG** workflow using **LangChain**, coordinating Search, Parser, Critique, and Slide Generation agents to automate investor presentation creation from scraped financial and company data
- Engineered **dynamic query-generation** and **prompt-planning** pipelines with overlapping Markdown chunking, semantic retrieval, and contextual refinement workflows, improving content population from 74.2% to 98% while reducing slide-planning latency by 49%
- Reduced slide-generation failures and low-content outputs by engineering an iterative **LLM critique** and repair pipeline from 70% to 90% and improving presentation completeness by 10–15% per iteration

LLMs are ACDs | *Python, CybORG, CAGE 4, OpenAI API, HuggingFace, Ollama*

Apr 2025 – Dec 2025

- Co-authored an **IEEE CAI 2025** paper developing the first framework for **LLM** and **RL** agents in **CybORG CAGE 4** multi-agent cybersecurity environment, enabling interpretable autonomous cyber defense simulations
- Collaborated within a 6-person research team to create an 8-bit communication protocol enabling coordination across agents (**LLM + RL**), supporting real time threat sharing and distributed defense across 4 network zones
- Benchmarked **OpenAI**, **Ollama**, and **Deepseek** against **RL** baselines across 5 attacking agents, applying behavioral analysis and reward-based evaluation to measure interpretability, action quality, and defensive performance
- Built **Python** adapters integrating **OpenAI**, **DeepSeek**, and **HuggingFace** APIs into **CybORG**, optimizing LLM action-selection workflows to support scalable 500-step adversarial simulations; RL agents achieved ~104× faster action selection (45s vs. 4,705s per episode)

PUBLICATIONS

LLMs are Autonomous Cyber Defense Agents | *IEEE Conference on Artificial Intelligence (CAI 2025)*

- IEEE Conference on Artificial Intelligence (CAI 2025) — arXiv:2505.04843